

BSCS 3214 Data Visualization- Lec/Lab

LESSON 2 – PRINCIPLES OF VISUALIZATION

January 19, 2026 M(10:00 AM)

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LESSON 2 – PRINCIPLES OF VISUALIZATION

Learning Objectives:

Knowledge (Cognitive Domain)

- Understand: Explain how human visual perception impacts the effectiveness of different chart types.
- Apply: Use pre-attentive attributes (color, size, position) to highlight critical data points in a visualization.
- Analyze: Analyze a given visualization to identify adherence to, or violations of, Melissa Anderson's design principles or Gestalt principles.
- Evaluate: Critique a data visualization's use of color theory and typography, justifying whether they aid or hinder the narrative flow.
- Create: Design a basic visualization pipeline that translates raw data into a compelling data story.

LESSON 2 – PRINCIPLES OF VISUALIZATION

Learning Objectives:

Skills (Psychomotor/Application Domain)

- Develop a color-blind accessible palette for a multi-series line chart.
- Implement visual hierarchy using typography and labeling to guide a viewer's eye through a complex dashboard.
- Construct a data visualization that follows a clear narrative flow (The Narrative Flow/Telling a Story).

LESSON 2 – PRINCIPLES OF VISUALIZATION

Learning Objectives:

Attitudes (Affective Domain)

- Value the importance of designing visualizations that are accessible to all users, including those with color blindness.
- Appreciate how thoughtful typography improves readability and professionalism in data storytelling.
- Respond to peer feedback on color choices and visual hierarchy with an openness to iteration.

Data Visualization - Course Outline

Lesson 2: Principles of Data Visualization

1. Visual Perception & Pre-attentive Attributes

- a. Melissa Anderson's Principles of Design
- b. Different Types of Visualization
- c. Challenges in data visualization
- d. Visualization Pipeline

2. Color Theory for Data

3. Gestalt Principles & Visual Hierarchy

4. Typography and Labelling

5. Telling a Story (The Narrative Flow)

Design Principles

1. Visual Perception & Pre-attentive Attributes

- **Visual perception** refers to the brain's ability to process and interpret visual information.
- **In data visualization, we leverage "pre-attentive attributes"**—visual elements detected by the brain almost instantly, before conscious thought is engaged.
- **Examples of Pre-attentive Attributes:**
 - Length: Easily compare bar lengths.
 - Color (Hue & Intensity): Spotting a red dot in a sea of grey dots.
 - Position: The most effective attribute; our eyes easily track points on an X/Y axis.
 - Size: Comparing circle areas.
 - Orientation: Instantly seeing vertical vs. horizontal lines.

Design Principles

A. Here are some of the key design principles for creating beautiful and effective data visualizations

Melissa Anderson's Principles of Design are a foundational set of guidelines used in data visualization and graphic design to bridge the gap between raw data and human cognition.

Unlike traditional art, which may prioritize expression, these principles focus on functional aesthetics—using design elements specifically to reduce "cognitive load" (the mental effort required to understand a chart) and ensure that the most important information is the most visible.

Design Principles

Core Principles:

1. Emphasis:

Draw the viewer's attention towards important data by using key visual elements. Emphasis is the component that is most related to when reading the nine principles of design. It is the key to being conscious of what is drawing the viewer's attention to the art. When thinking about the art design of data visualization is either drawing the viewer to that point of emphasis or how they are being distracted or drawn elsewhere.

Design Principles

Core Principles:

2. Balance:

- A design is said to be balanced if key visual elements as such as color, shape, texture, and negative space, are uniformly distributed. And it is important to remember that non-data elements, such as a logo, title, and caption, can affect the balance of the display.

Design Principles

Core Principles:

2. Balance:

- Balance refers to the distribution of visual "weight" across a dashboard or page.
- The Goal: To make the visualization feel stable and organized, rather than lopsided or overwhelming.⁵
- Application: Symmetrical Balance: Centering elements (often used for formal reports).
 - Asymmetrical Balance: Balancing a large, heavy chart on the left with several smaller, lighter text boxes or KPIs on the right.

Design Principles

Core Principles:

3. Unity (Harmony):

Unity is the sense that all elements on a dashboard belong together.

- Unity across the design will happen naturally if all other principles are implemented.
- The Goal: To prevent the "Frankenstein effect," where a dashboard looks like a collection of random charts pasted together.
- Application: Use a consistent color palette, the same font family across all labels, and consistent spacing (margins) between charts.

Design Principles

Core Principles:

3. Proportion (Scale):

If a person is portrayed next to the house, the house will appear larger in data visualization; the proportion can indicate the importance of the dataset, along with the actual relationship between numbers.

Proportion can be subtle, but it can go a long way to enhancing a viewer's experience and understanding of the data. The danger of proportion, though, is that it can be easy to deceive people subconsciously.

Design Principles

Core Principles:

4. Proportion (Scale):

Naturally, images will have a greater impact on how our brains perceive the dashboard or visualization.

For example, someone can change the scale of a graph or images to inflate their results, and even if they write the numbers next to it, the shortcut many people will take is to interpret the data based on the image.

Design Principles

Core Principles:

4. Proportion (Scale):

- Proportion is the relationship between the sizes of different elements.
- The Goal: To ensure that the visual size of a chart matches its importance or the magnitude of the data it represents.
- Application: Don't give a minor "secondary" metric a massive pie chart while the "primary" KPI is tucked away in a tiny corner.

Design Principles

Core Principles:

5. Rhythm (Repetition)

- **Rhythm** is created by repeating visual elements to create a sense of movement or organized structure.
- The Goal: To help the user "learn" how to read your dashboard quickly.
- Application: If you use a blue header for one chart, use a blue header for all of them. If the X-axis is at the bottom of one bar chart, keep it at the bottom for the rest. This creates a predictable "rhythm" for the eye.

Design Principles

Core Principles:

6. Contrast:

- **Contrast** is the most powerful tool for showing difference.
- The Goal: To clearly distinguish between different categories, or between the data and the background.
- Application: Use high-contrast colors (like black text on a white background) for readability. Use contrasting shapes (circles vs. squares) to differentiate data series in a scatter plot.

Design Principles

b. Different Types of Visualization

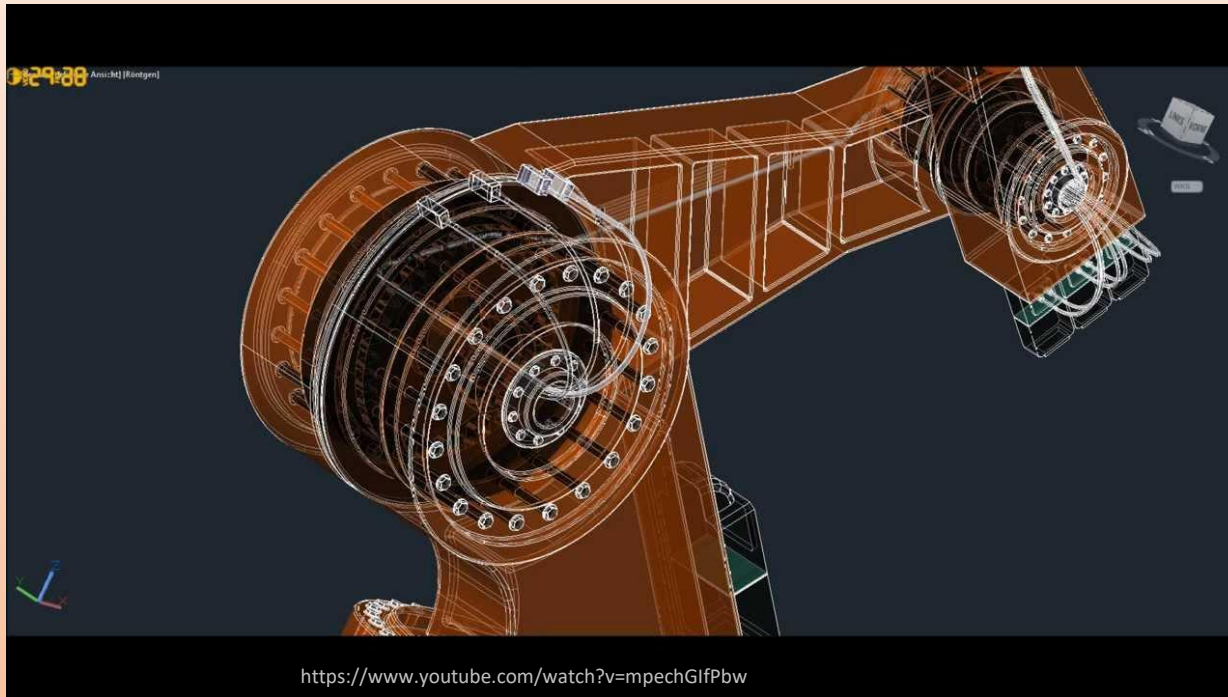
The choice of chart depends entirely on the data relationship you want to show:

- Comparison: Bar charts (vertical/horizontal) are the standard for comparing discrete categories.
- Relationship: Scatter plots or bubble charts show how two or three variables interact.
- Distribution: Histograms or density plots show how data is spread out.
- Composition: Stacked bar charts or tree maps show parts of a whole.

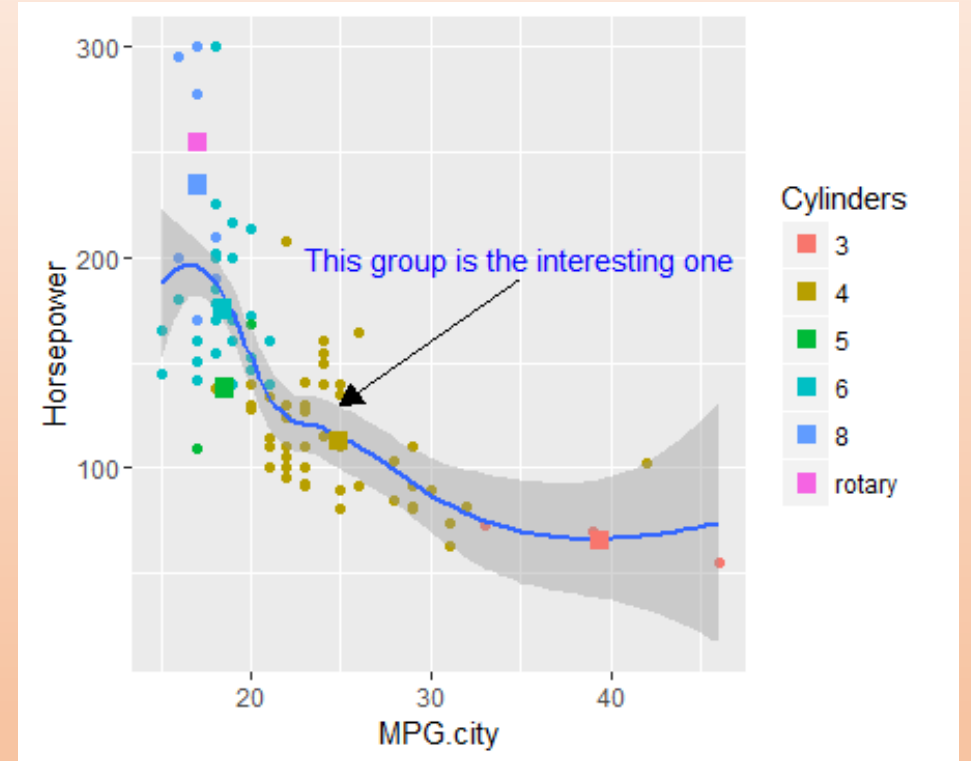
Different Types of Visualization

- **Visualization = information visualization**
 - Data → Visualization → Analysis
- Related concepts
 - **Computer graphics**: Data is not necessarily present, analysis is not normally assumed
 - Example: Computer games
 - **Scientific visualization**: Similar to information visualization, often engineering data, statistical/machine learning analysis is normally not assumed
 - Example: Industrial robots

Different Types of Visualization



Scientific Visualization

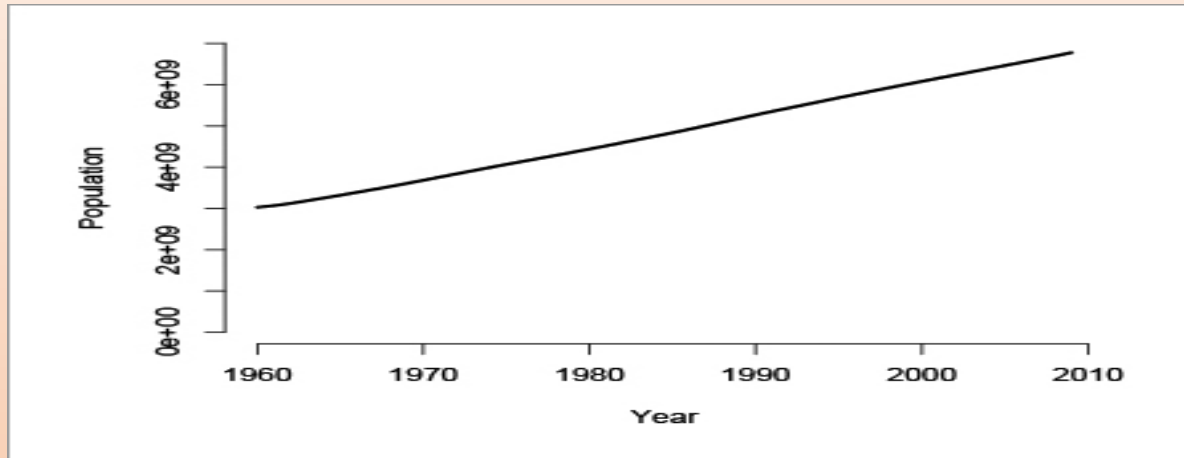


Information Visualization

Challenges in Data Visualization

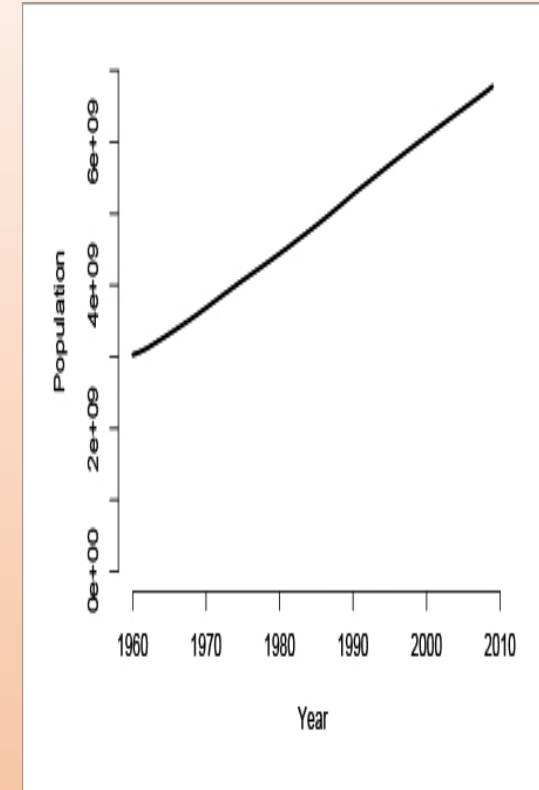
- **Cognitive Overload:** The primary challenge is giving the user too much information to process at once (high Extraneous Load).
- **Accessibility:** Ensuring visualizations are readable by all users, particularly those with color vision deficiencies (CVD).
- **Misinformation (The Lie Factor):** Creating visualizations that are technically accurate but visually misleading (e.g., using a truncated Y-axis).

Why is Visualization Important?



Decision here: Population does not increase so much, no intervention needed

Visual perception problem



Decision here: population increases quickly, intervention is required

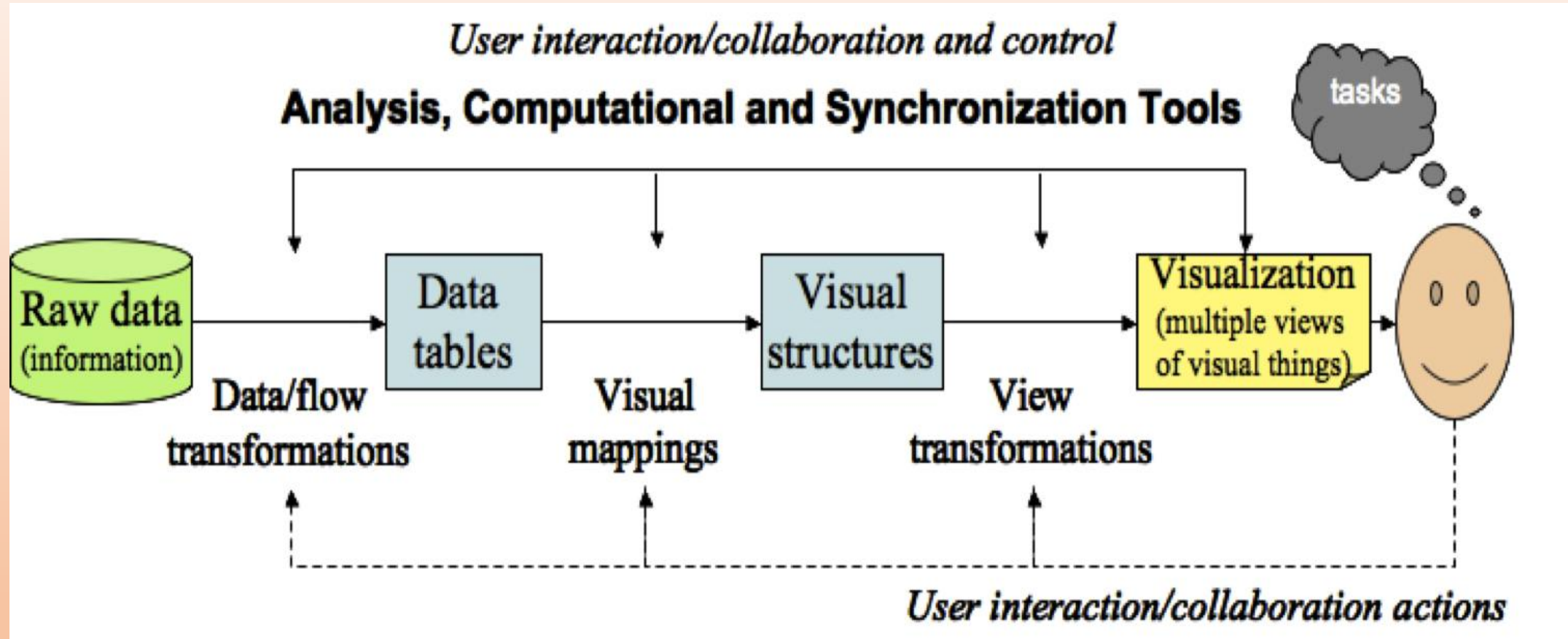
Visualization Aims

- Visualization for **exploration**
 - Clusters
 - Trends
 - Anomalies
 - ...
- **Confirmatory** visualization
 - Example 1: Perform linear regression, analyse residuals → was linear regression reasonable
 - Example 2: Discover clusters by K-means, visualize clusters → are they clusters actually?
- Visualization for **presentation**

Visualization Pipeline

Visualization Pipeline, formalized by Card, Mackinlay, and Shneiderman (1999), is a foundational reference model that describes the systematic process of converting raw data into a visual representation that amplifies human cognition. It represents a series of data-driven transformations designed to help users extract insights and form mental models.

Visualization Pipeline



Key ingredient: mapping data columns to visual structures (aesthetics)

Visualization Pipeline

Core Stages of the Pipeline

The model semantically delineates the primary phases a dataset undergoes from its raw format to a final rendered image:

- Data Transformations (Raw Data → Data Tables):
 - This initial step cleans, reshapes, and summarizes source datasets into organized data tables.
 - Common tasks include filtering, normalization, handling missing values, and dimensionality reduction to prepare the data for analysis.

Visualization Pipeline

Visual Mapping (Data Tables → Visual Structures):

- Considered the "heart" of the process, this stage links abstract data attributes to visual structures.
- Designers map data entities to visual marks (such as dots or bars) and assign graphical properties like spatial substrate (position), size, color, and shape.

Visualization Pipeline

View Transformations (Visual Structures → Human Views):

- The visual structures are embedded into actual views for screen display.
- This involves setting interactive parameters like scaling, clipping, and navigation (panning/zooming) to allow the user to explore the data.

Visualization Pipeline

Key Features of the Model

- **Cognitive Amplification:** The purpose of this pipeline is not merely to create "interesting pictures" but to act as a tool that assists visual reasoning and pattern recognition beyond simple data transfer.
- **Human Interaction:** The model is highly interactive; users can intervene at any stage of the pipeline to alter the resulting visualization or change the view parameters, facilitating deeper interpretation and insight generation.
- **Loop-back Process:** In dynamic graph visualizations, the pipeline may be revisited multiple times to analyze new data or recompute layouts for nodes and edges.

Color Theory for Data

2. Color Theory for Data

- The strategic application of color principles to encode information, clarify complex insights, and guide human visual perception.
- Unlike traditional art where color may be purely decorative, in data visualization, it serves a functional purpose: transforming abstract numbers into a readable, impactful narrative.
- Effective use of color leverages the brain's "pre-attentive processing," allowing viewers to spot trends or anomalies instantly before even consciously reading the chart.

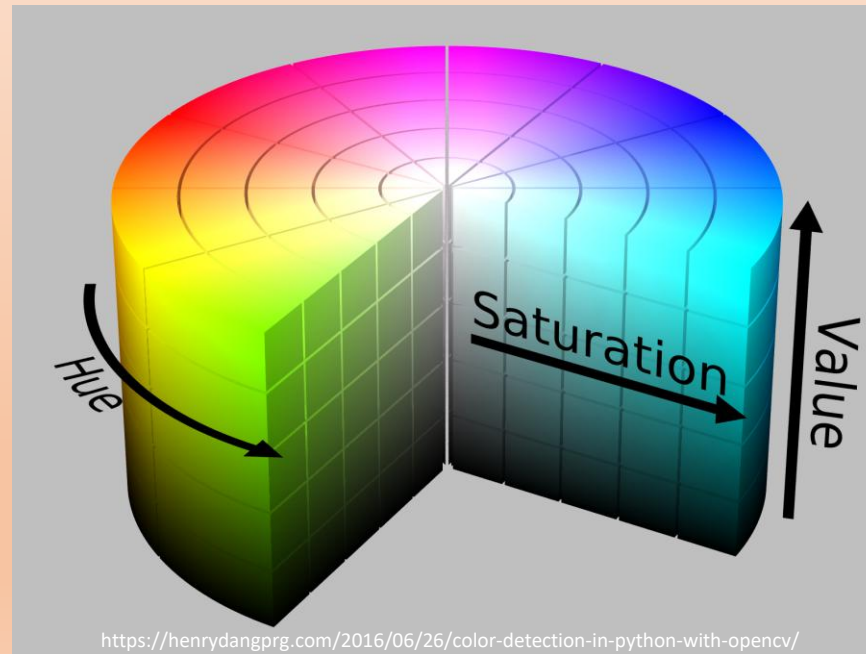
Color Theory for Data

2. Color Theory for Data

- Color should be functional, not decorative.
- Hue: The pure color (Red, Green, Blue).
- Saturation (Chroma): The intensity or purity of the color.
- Luminance (Brightness): The lightness or darkness of the color.
- Accessibility: Use color palettes that are CVD-friendly (e.g., blue-orange palettes) and use secondary encodings (patterns, labels) to ensure all information is available even without color.

Colors

- Color= hue + saturation + value (lightness)
- 8% of males are color deficient → what are good colors?



Color Theory for Data

Core Dimensions of Data Color

To apply color theory effectively, designers break each hue into three separate channels, often referred to as the HSL model:

- **Hue:** The actual color (e.g., Red, Blue, Green). It is most effective for distinguishing unrelated categorical (qualitative) groups.
- **Saturation (Chroma):** The intensity or vividness of a color. Saturated colors grab attention but can quickly overwhelm the viewer if overused.
- **Luminance (Value):** The perceived lightness or darkness. Luminance is the most critical attribute for showing sequential (ordered) data, as the brain naturally associates darker or more intense shades with higher values.

Color Theory for Data

2026 Standards for Data Palettes

Choosing the right palette depends entirely on the nature of the underlying data:

- **Qualitative (Categorical):** Distinct hues for unordered data (e.g., different company regions). Limit these to 6–12 colors to avoid cognitive overload.
- **Sequential:** A single hue varying in luminance for ordered data (e.g., population density).
- **Diverging:** Two contrasting hues that meet at a neutral midpoint (e.g., profit vs. loss, or temperature above/below average).

Color Theory for Data

Functional Roles of Color

- **Emphasis:** Using a bright accent color against neutral tones to highlight a specific data point, such as an outlier or the current year's performance.
- **Grouping:** Applying the same color to related data points across different charts to maintain a consistent visual hierarchy.

Color Theory for Data

Functional Roles of Color

- **Semantics:** Aligning color choices with established cultural or industry norms, such as red for "danger" or green for "growth," to make the visualization intuitive.
- **Accessibility:** Ensuring designs are readable for the 1 in 12 men with Color Vision Deficiency (CVD) by avoiding red-green combinations and using simulators like Coblis.

Color Theory for Data

Gestalt Principles & Visual Hierarchy

- **Gestalt psychology** describes how humans group similar visual elements automatically. Using these principles reduces Extraneous Cognitive Load:
- Proximity: Objects close together are perceived as a group.
- Similarity: Objects that look similar (color, shape) are perceived as a group.
- Continuity: The eye naturally follows the smoothest path, line, or curve.

Color Theory for Data

Gestalt Principles & Visual Hierarchy

- Closure: The mind fills in missing visual information to complete a shape.
- Visual Hierarchy: The strategic use of size, weight, and color to show relationships. The most important data point should be the most visually dominant thing on the screen.

Color Theory for Data

Telling a Story (The Narrative Flow)

- A powerful visualization moves beyond a collection of charts and becomes a guided story. This creates Germane Cognitive Load (the good kind of effort).
- The Narrative Arc:
 - The Hook (Overview): Grab attention with a high-level summary (e.g., "Global sales increased 45%").
 - The Conflict (Zoom & Filter): Dive into the details that explain the hook (e.g., "But only in the Asia region").
 - The Resolution (Insight): The final reveal or call to action (e.g., "We must invest heavily in Asia operations").

Color Theory for Data

Telling a Story (The Narrative Flow)

- Technique: Use annotations and filtering to guide the viewer from Step 1 to Step 3 seamlessly.

The Role of Perception

- Human visual system has limitations
- These limitations may lead to wrong/incomplete analysis of graphs
- Understanding how we see → better displays
- Misleading graphics need to be avoided

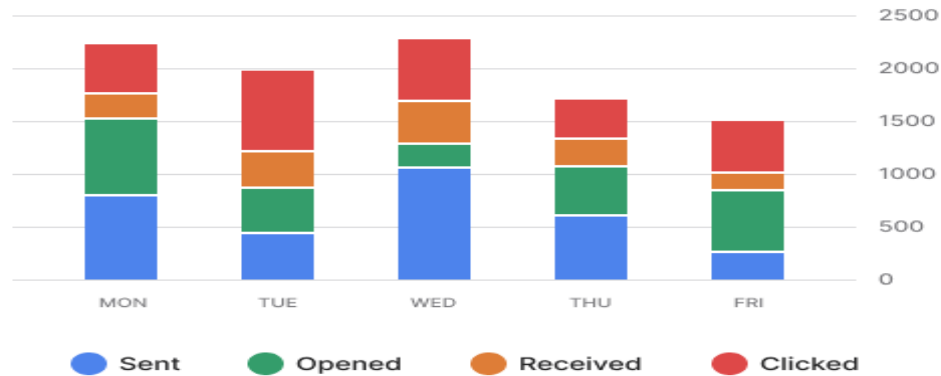
Bar Chart Example

- In this example, the red and green bars have very similar levels of brightness and saturation. For users with color vision deficiencies (CVD), such as deuteranopia (red-green color blindness), these bars may appear almost identical in hue, making it impossible to distinguish between categories without additional visual cues.

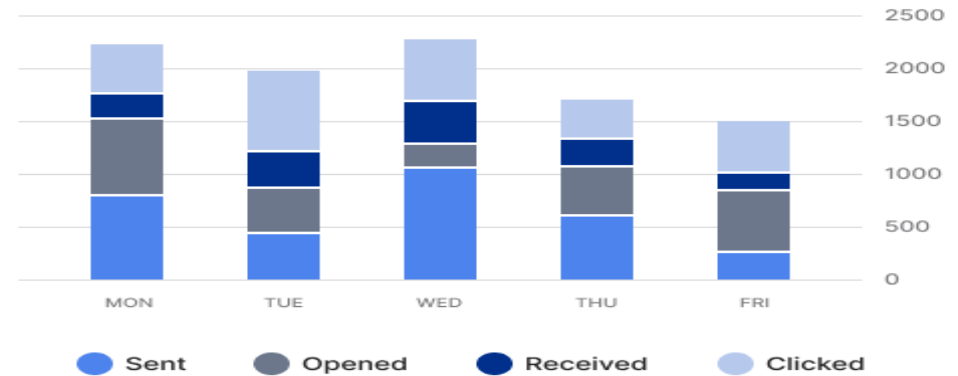
The Role of Perception

✗ Multi-Color Marks

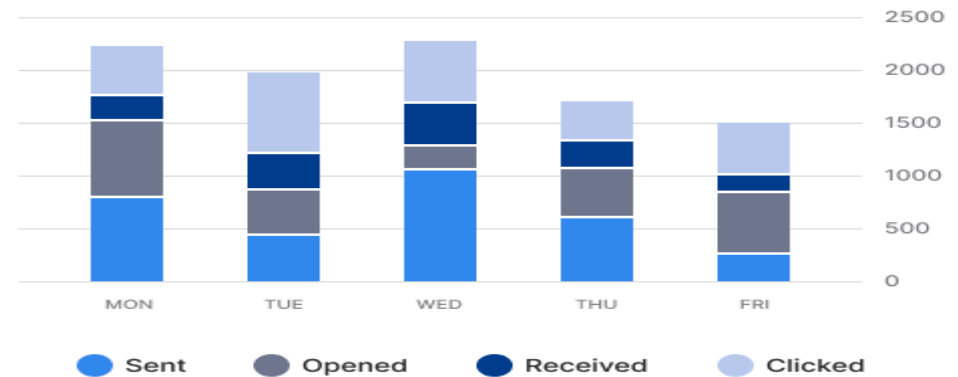
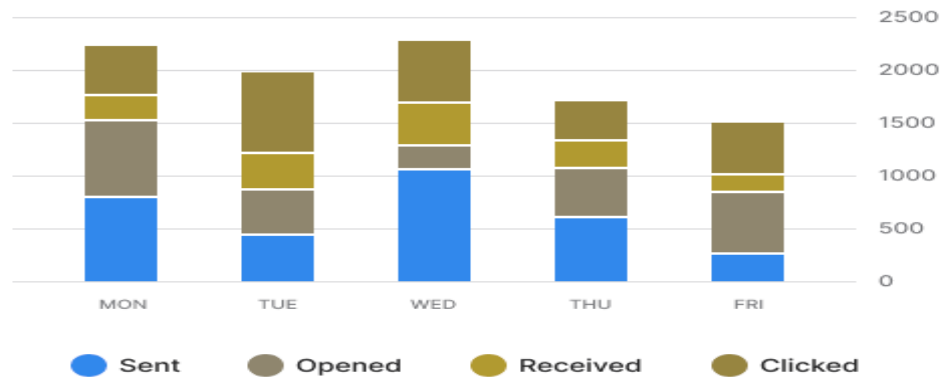
Normal Vision



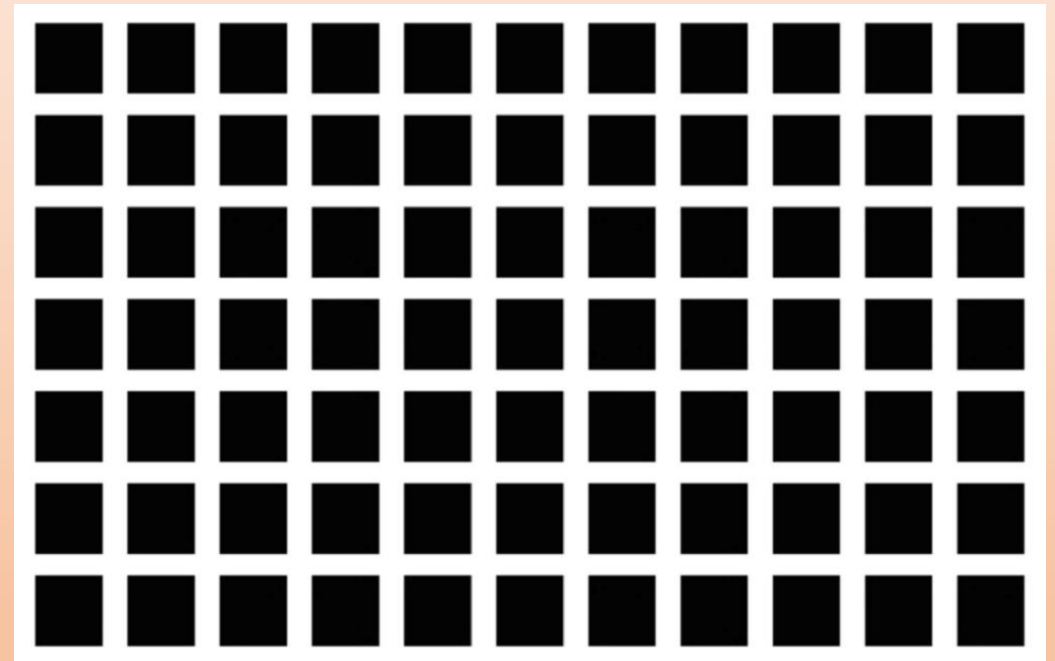
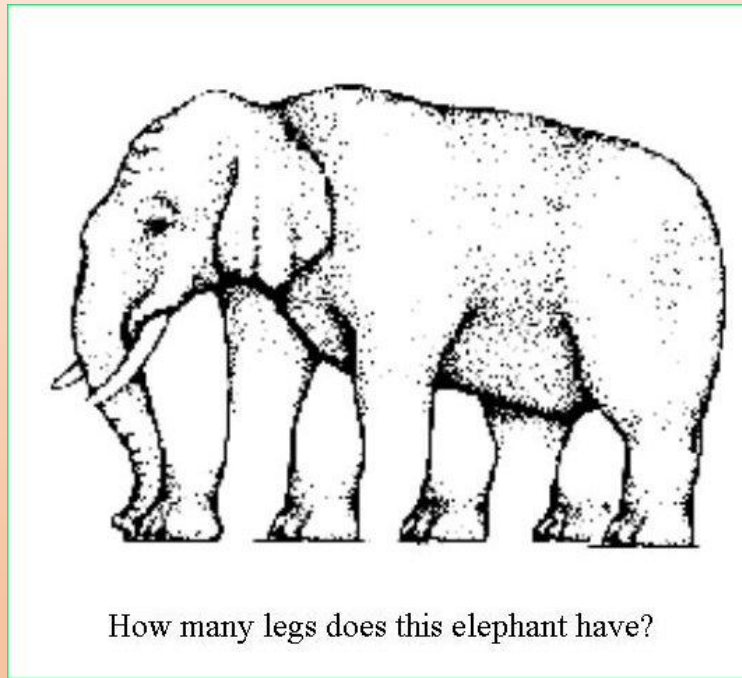
✓ Multi-Shade Marks



Seen with Red-Green Color Blindness



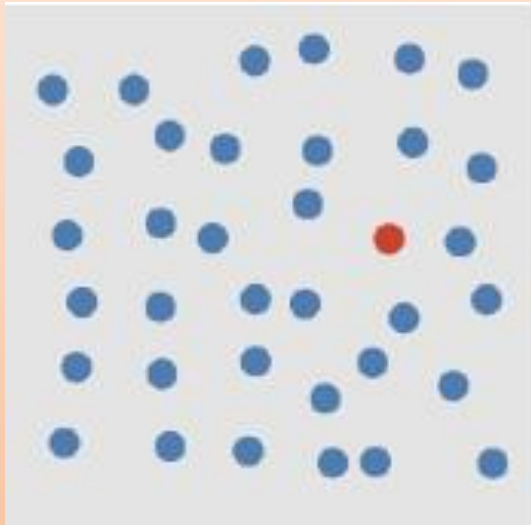
Illusions



<http://www.ilusa.com/gallery/elephant-illusion.jpg>

Pre-attentive Processing

- Certain aesthetics are fast to process



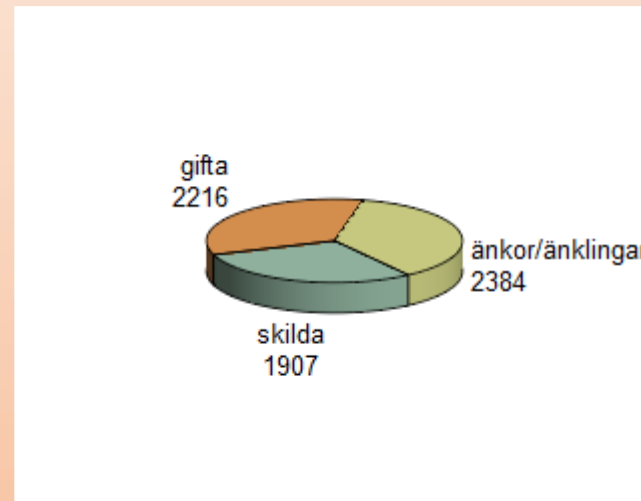
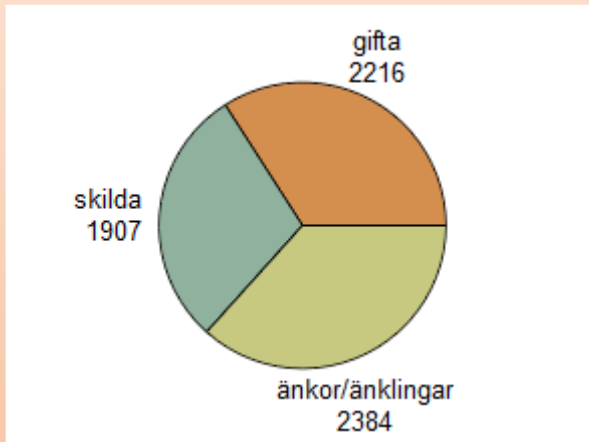
How quickly can you identify a red dot?



How quickly can you identify a square of right-handed Rs?

The role of perception

- How can this affect analysis?



Data pre-processing

- Viewing raw data is often preferred
- Sometimes some preprocessing is needed
- **Missing values and Data cleaning**
 - Discard the bad record → may remove almost all data
 - Assign sentinel value
 - Column mean imputation
 - Nearest neighbor imputation
 - Other imputations

Data preprocessing

- Normalization

- Converting column to range $[0,1]$. Useful in for ex. color mapping
- Centering and scaling 0/1
- Nonlinear transformations: log, sqrt

- Segmentation

- Split data according to some column

Data preprocessing

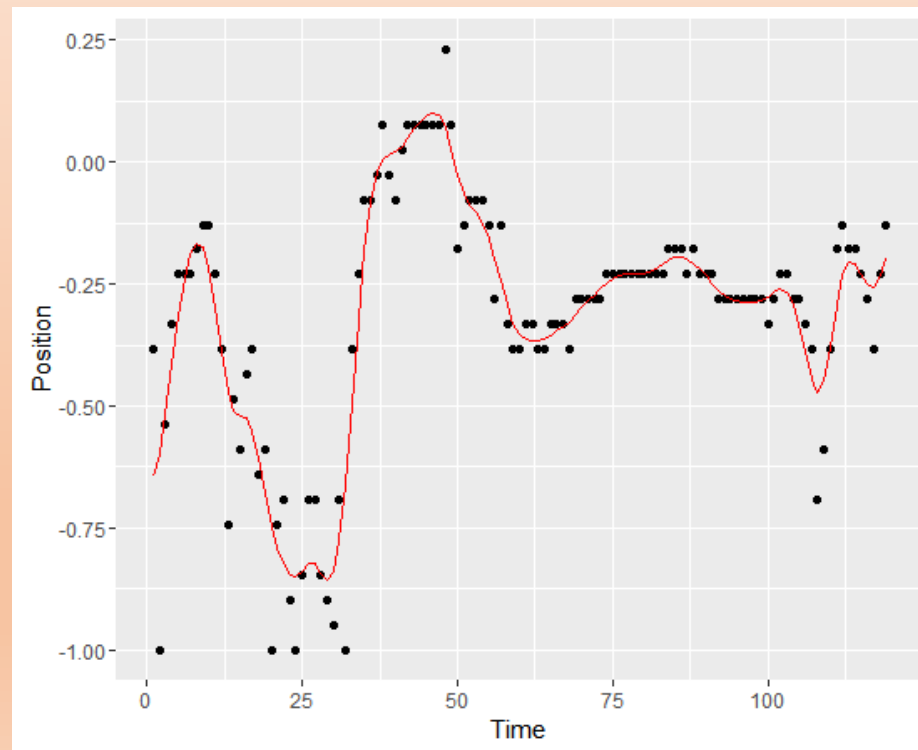
- Sampling, subsetting and expanding
 - Random sampling reduces size of data and facilitates overplotting (for ex. scatterplots)
 - Interpolation: linear (one dimension), bilinear (two dimensions), nonlinear. Select necessary amount of interpolation points.
- Dimension reduction
 - PCA
 - MDS
 - Other techniques (ex. ICA, Autoencoders), welcome to **Machine Learning** course..

Data preprocessing

- Mapping nominal dimensions to numbers
 - Random mapping should never be done unless intrinsic ordering is present
 - Use other numeric variables to measure in the data to measure “closeness” of values in the nominal variable
 - Correspondence analysis
- Aggregation and summarization
 1. Grouping observations
 2. Computing summary statistics per group

Data preprocessing

- Smoothing and filtering
 - Replace original values with a smoothed versions



Software

Commercial:

- [SAS and SAS JMP](#) – environment. Special visual tools are available (JMP), require separate license. Well documented. Good even for large sets. [SAS Enterprise Guide](#) has many visual static tools
- [Spotfire](#) – Many static and interactive visualization tools
- [Tableau](#) – Many static and interactive visualization tools
- [InfoScope](#) – visualizing maps, interactive visualizations

Free:

- [R](#) – programming language. Set of packages is constantly updated. A lot of statistical tools (even the newest methods) Badly documented
- [Plotly](#) – a tool for interactive and dynamic graphics, R interface available
- [Shiny](#) – a tool for R-based web applications using graphics
- [GraphViz](#) – visualization of graph data, coding needed
- [Jigsaw](#) – Text analysis

Software

Tools for the web (used by web designers):

- [ActionScript](#)
- [JavaScript](#)
- [Prefuse](#)
- [VTK](#)

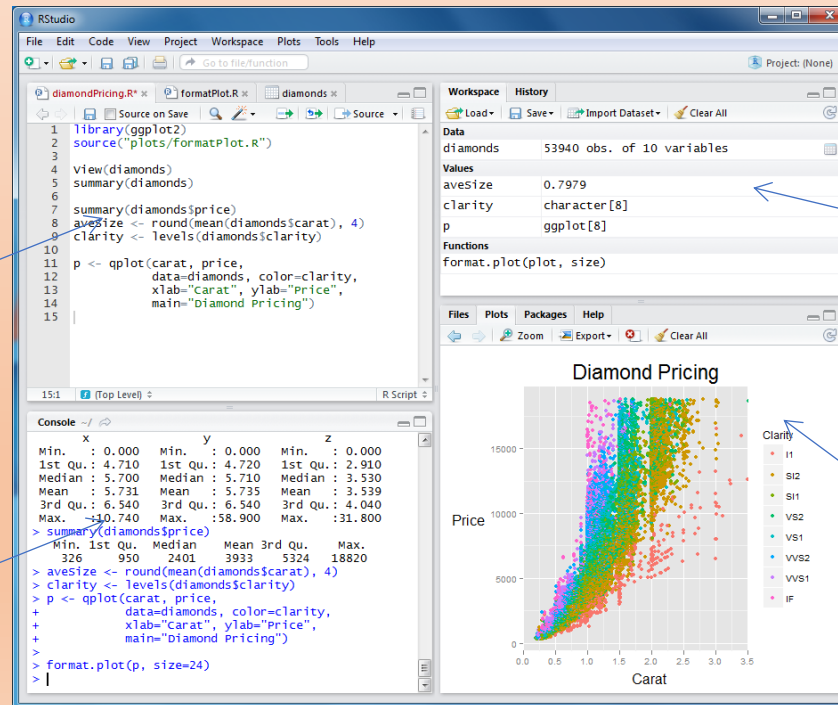
... much more references given in the course book

Course software

- Install R: <http://www.r-project.org/>
- Install RStudio: <http://rstudio.org/>

Program

Execution
console

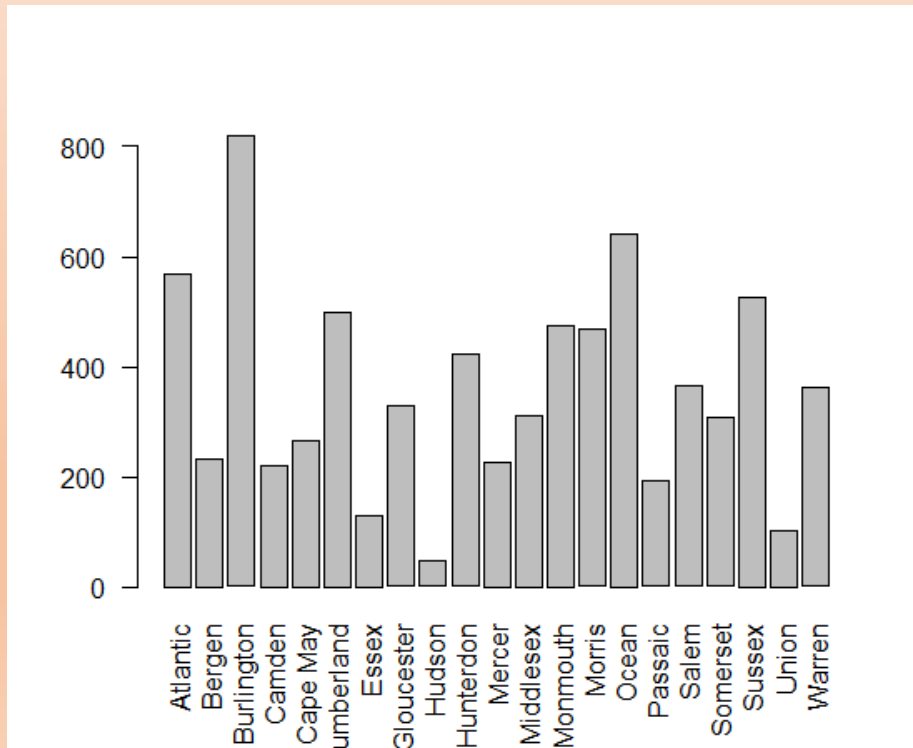


Workspace

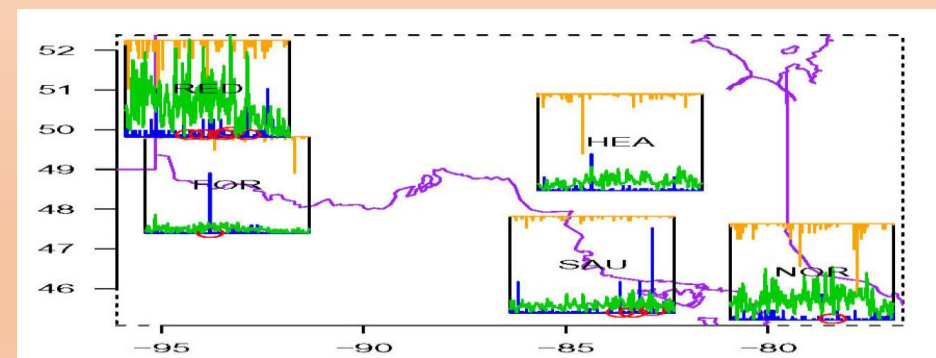
Plots

Graphical tools in R

- R base: basic plotting, not publication quality



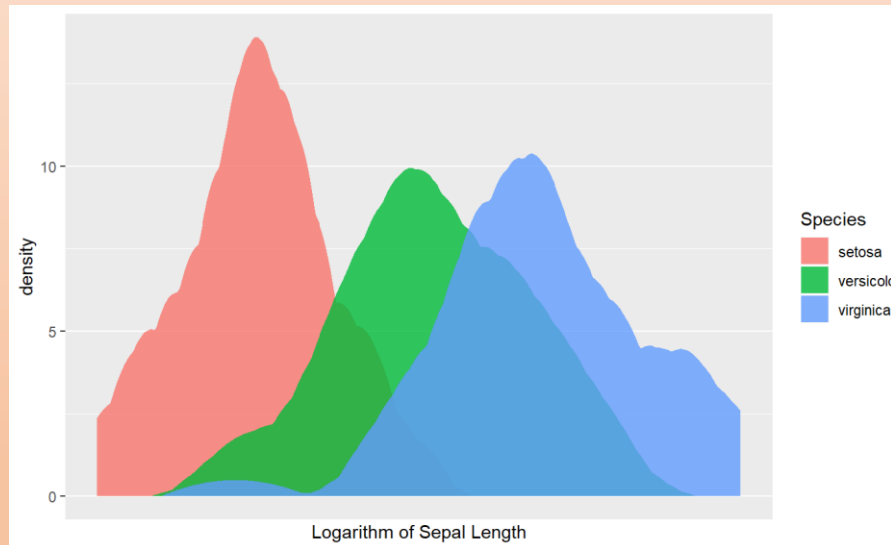
Grob package: low-level plotting tools, new types of plots



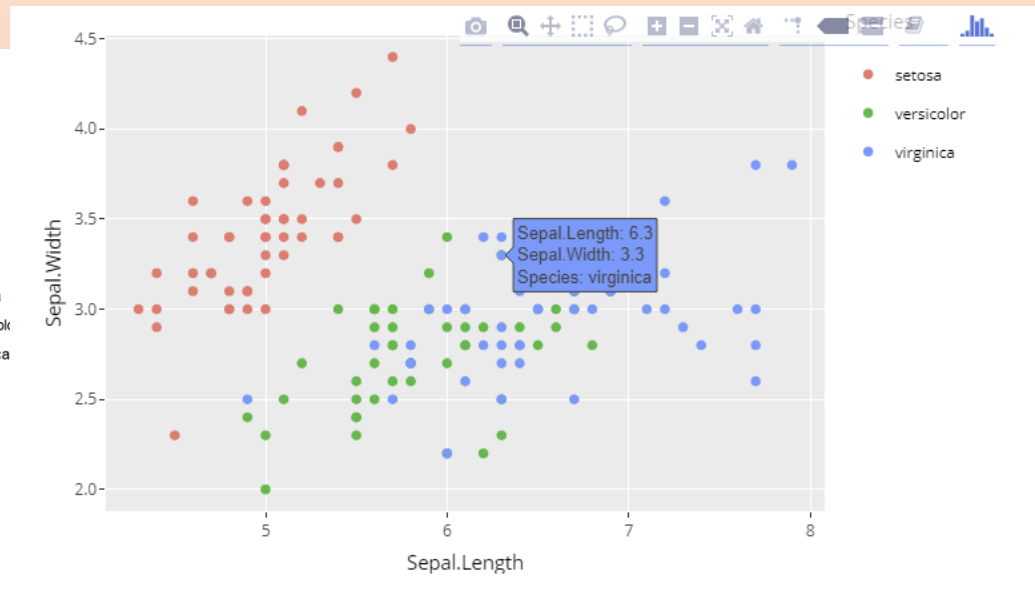
Zhou, Lutong, and W. John Braun. "Fun with the r grid package." *Journal of Statistics Education* 18.3 (2010).

Graphical tools in R

Ggplot2 package: based on **grammar of graphics**, close to publication quality



Plotly package: Ggplot2 + interactivity

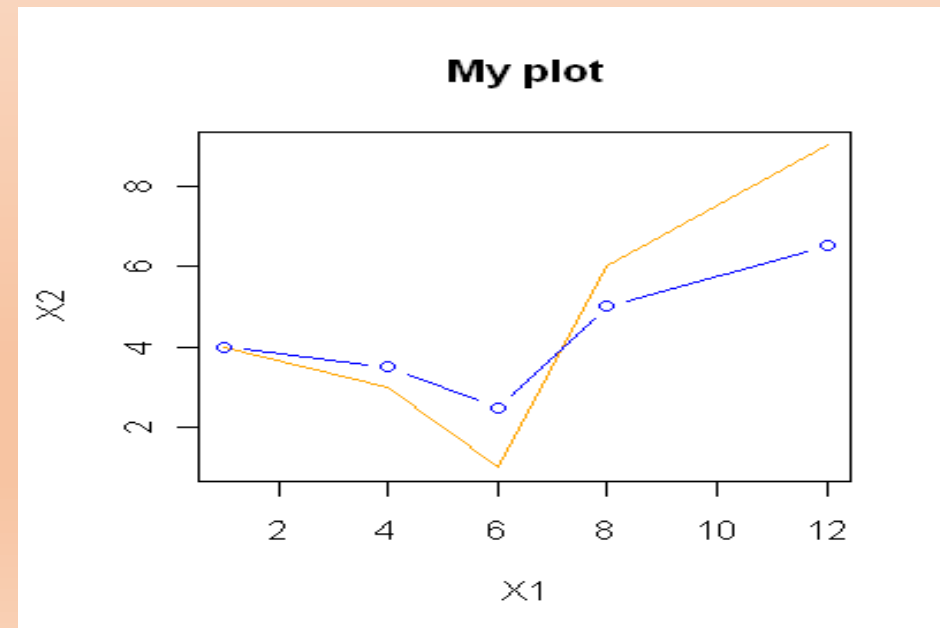


Graphical procedures

Base R graphical procedures:

- `plot(x,..)` plots time series
- `plot(x,y)` scatter plot
- `plot(x,y)` followed by `points(x,y)` plots several scatterplots in one coordinate system
- `hist(x,..)` plots a hitogram
- `persp(x,y,z,...)` creates surface plots
- `cloud(formula,data..)` creates 3D scatter plot

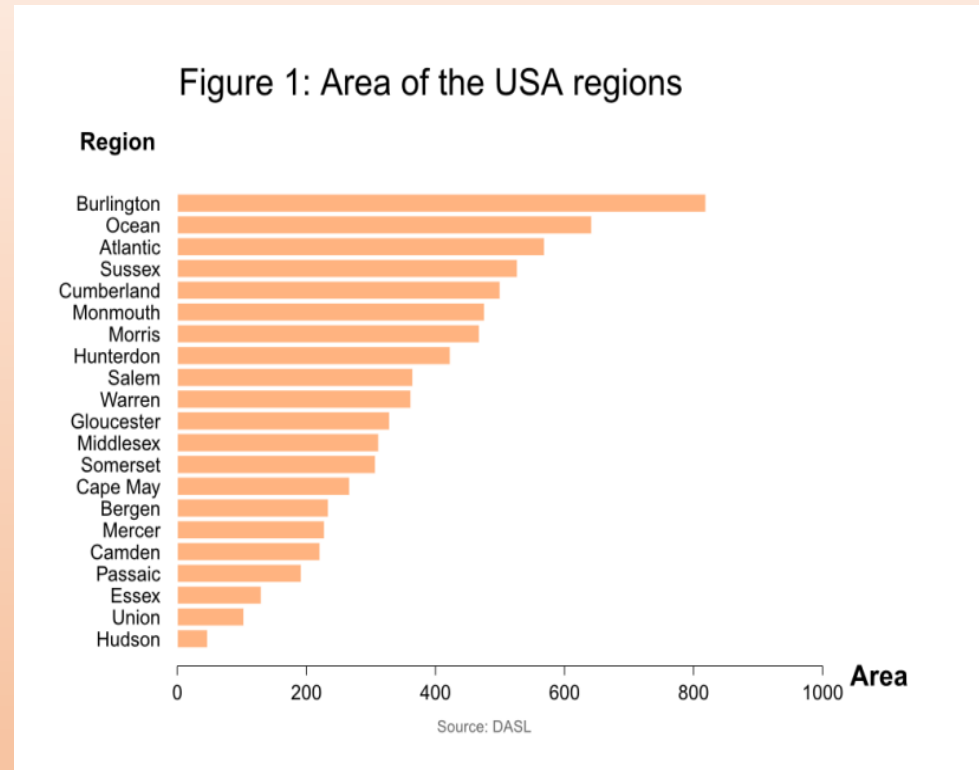
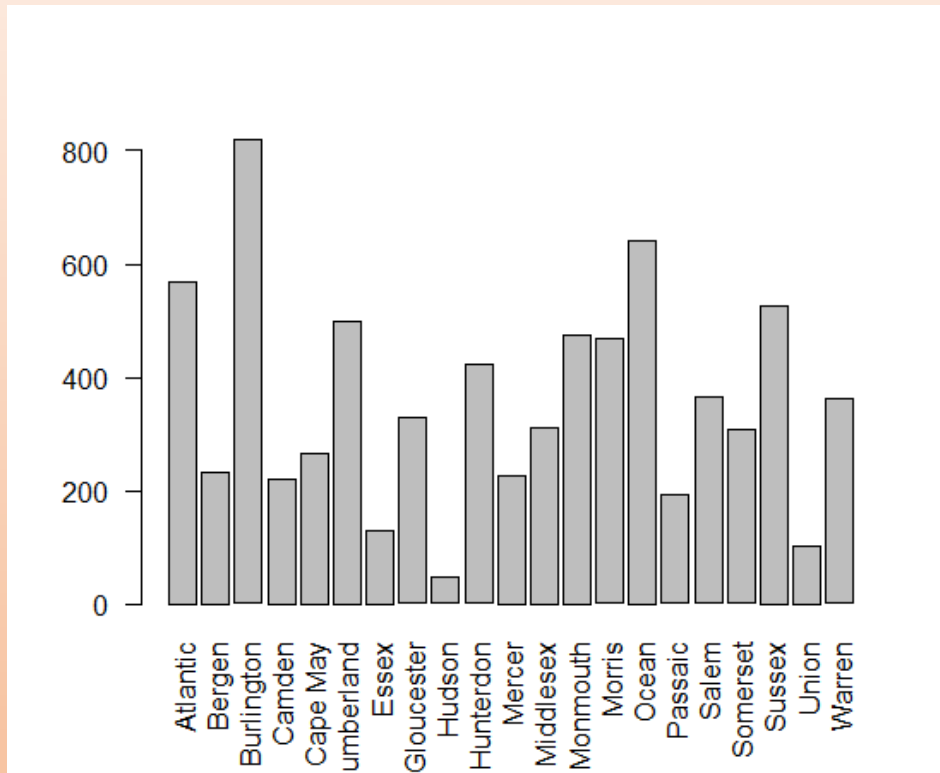
```
x<-c(1,4,7 8, 12);  
y<-c(4,3,1,6,9);  
  
plot(x,y, type="l", col="orange",  
      main="My plot", xlab="X1", ylab="X2");  
points(x, y/2+2, type="b", col="blue");
```



Publication quality graphics

- Visualization for **exploration**
 - Default settings
- Visualization for presentation for **publication**
 - Higher quality graphics is required
 - Improve the graph quality in the software (often requires quite a bit of programming)
 - Use postprocessing tools, such as Inkscape or Adobe Illustrator

Publication quality graphics



Plot processed in
Inkscape

Example: Compare two plots and state what is improved in the second plot.

Making publication quality graphics

- Install **Inkscape**
 - <http://inkscape.org/>
 - Inkscape is an open source, SVG-based vector drawing program
 - file format that Inkscape uses is compact and quickly transmittable over the Internet.
 - Vector graphics: image is defined in terms of lines, not pixels
 - Benefit: can be enlarged without loss of picture quality
- 1. Save your R plot as PDF and import it to Inkscape
- 2. Make changes and export your plot as a PNG-file or save it as PDF.

Bitmap image and vector image (enlarged)



Inkscape

- Menu bar (**Important:** File, Object, Path, Text, View→Grid)
- Command bar (Zoom to fit page, Edit object's colors)
- Tool controls
- Tool box (Select and transform, zoom, Text, Eraser, Fill)
- Color palette
- Status bar

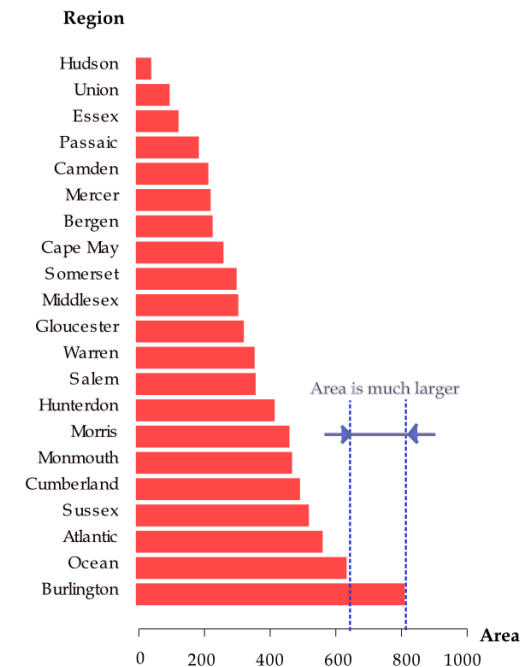
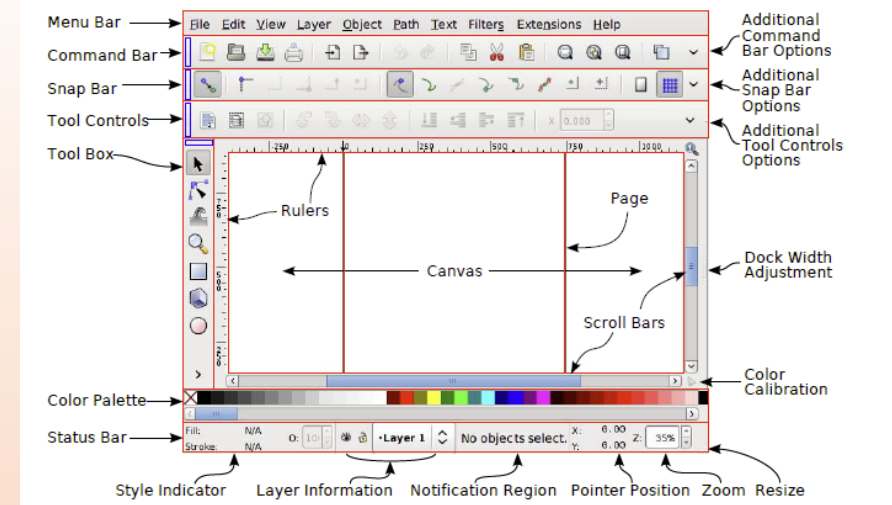


Figure 1: There seems to be one outlying region

Activity:

Activity: The "Accessibility" Color Audit

- Scenario: A 2026 Smart City initiative requires all public-facing dashboards to be WCAG 2.1 AA compliant. You have been given a standard bar chart that uses an inaccessible red/green/blue color palette. Your task is to redesign the colors to be universally readable while maintaining visual hierarchy.

Activity:

The "Accessibility" Color Audit

- Objectives:
 - Identify which colors fail standard color-blindness checks (Deuteranopia, Protanopia).
 - Develop a new palette using a Color Theory tool (e.g., Adobe Color, ColorBrewer).
 - Apply Melissa Anderson's principle of "Similarity" to group related data points effectively.
- Deliverables:
 - A screenshot of the "Before" (inaccessible) chart.
 - A screenshot of the "After" (accessible) chart.
 - A one-page justification of your color choices and how they leverage Pre-attentive Attributes to guide the eye.

Activity:

Lab Activity 2: The "Accessibility" Color Audit

- **Goal:** Redesign an inaccessible chart to meet 2026 accessibility standards (WCAG 2.1 AA).

Step-by-Step Workflow:

1. Selection & Analysis:

- Open the provided "Inaccessible RGB Chart."
- Identify the "Pain Points": Does it rely solely on color? Is there a legend that is hard to read?

Activity:

Lab Activity 2: The "Accessibility" Color Audit

3. Color Blindness Simulation:

- Use the Coblis (Color Blindness Simulator) or the built-in CVD simulator in Adobe Photoshop (View > Proof Setup > Color Blindness).
- Take a screenshot of how the chart looks to someone with *Deuteranopia*.

4. Palette Development:

- Go to Adobe Color and select the "Accessibility Tools" tab.
- Generate a High-Contrast Palette (e.g., Blue/Orange or Sequential Blues) that passes the "Color Blind Safe" check.

Activity:

Lab Activity 2: The "Accessibility" Color Audit

5. Redesign (The "Double Encoding" Technique):

- Apply your new colors to the bars.
- Add Patterns: Use the "Fill Effects" in your design tool to add stripes or dots to specific bars so they are distinguishable even in grayscale.

6. Direct Labeling:

- Remove the legend. Place the category names directly above each bar to reduce Extraneous Cognitive Load.

7. Final Export: Save as a PDF/PNG and write your 1-page justification.